INCH-POUND
MS21433F
5 October 2011
SUPERSEDING
MS21433E
4 September 1991

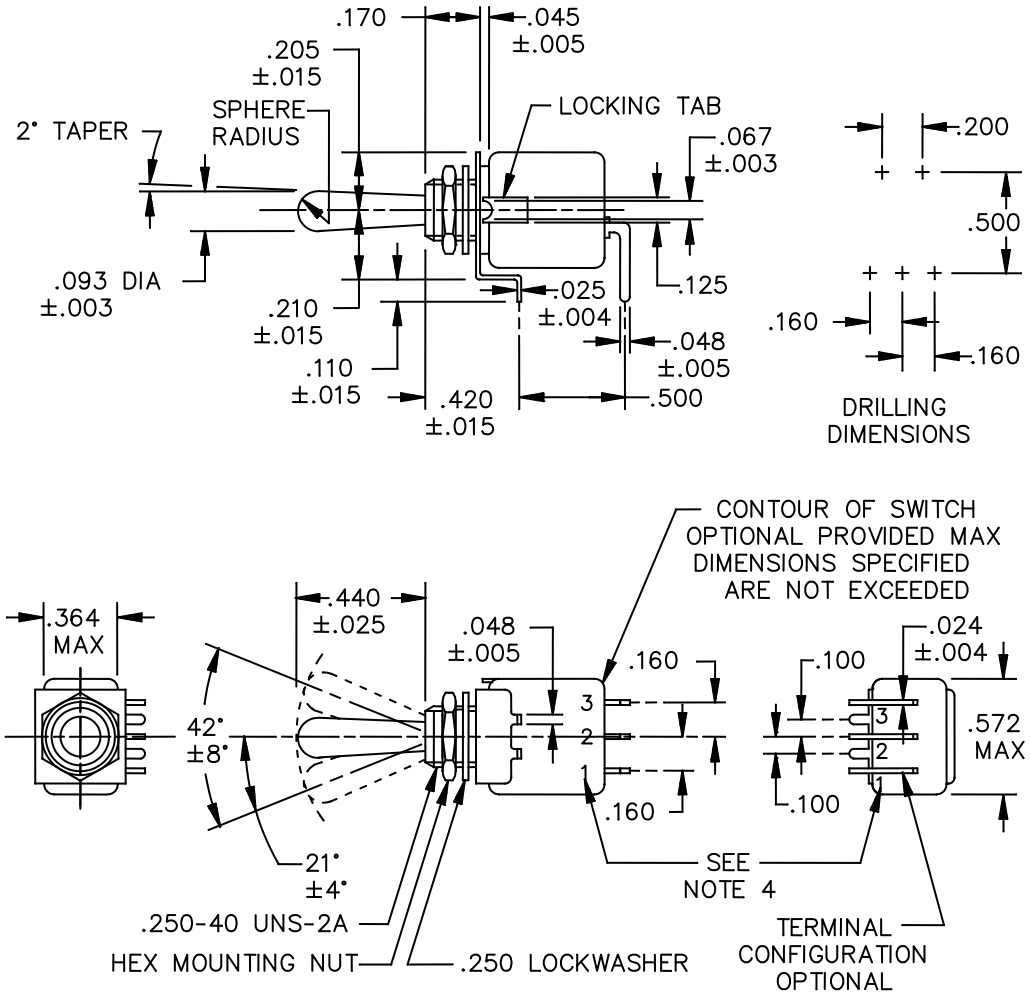
DETAIL SPECIFICATION SHEET

SWITCH, TOGGLE, POSITIVE BREAK, MINIATURE, TOGGLE SEALED, SINGLE POLE, PRINTED CIRCUIT BOARD TERMINALS, RIGHT ANGLE MOUNT

Inactive for new design after 5 October 2011

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification sheet and MIL-DTL-8834.



-FIGURE 1. Dimensions and configuration.

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Inches	mm	Inches	mm	Inches	mm		
.003	0.08	.045	1.14	.125	3.18	.250	6.35
.004	0.10	.048	1.22	.160	4.06	.364	9.25
.005	0.13	.067	1.70	.170	4.32	.420	10.67
.015	0.38	.093	2.36	.200	5.08	.440	11.18
.024	0.61	.100	2.54	.205	5.21	.500	12.70
.025	0.64	.110	2.79	.210	5.33	.572	14.53

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) on decimals and ±5° on angles.
4. Terminals need not be marked. Terminal identification is shown for reference purposes only.

FIGURE 1. Dimensions and configuration - Continued.

REQUIREMENTS:

For hardware detail specifications, see appendix of MIL-DTL-8834.

In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Shock: Method I and method II (high impact). The switches shall be rigidly mounted to the shock test equipment by means of their terminals and mounting tab. The switch shall be electrically and mechanically operative at the conclusion of the test and there shall be no mechanical transfer during the test.

Terminals: Gold plated.

Weight: .0165 pound maximum (6.15 grams).

Strength of terminals: Terminals shall be subjected to a five-pound pull for 5 to 10 seconds with the long dimension of the terminal contact vertical. The force shall be applied to the terminal at a point .06 inch (1.5 mm) below the bottom of the switch base, and in a direction parallel to the axis of the toggle lever in its center position. Two outboard terminals shall be tested. However, a terminal shall be tested only once in one direction.

Altitude: 50,000 feet.

Strength of actuator. Lever pivot and lever stop 6 pounds.

Mounting bracket: Brass, tin plated, or coated.

For hardware detail specifications, see appendix of MIL-DTL-8834.

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TABLE I. Detail requirements.

MS dash no	Circuit made between terminals as indicated with the toggle lever in these positions: 1/			Current capacity in amperes						Low current level switching 5 mV
				Resistive load			Inductive load			
				28 volts dc	115 volts		28 volts dc	115 volts		
	Opposite locking tab side	Center position	Locking tab side 4/		60 Hertz ac	400 Hertz ac		60 Hertz ac	400 Hertz ac	
-211	on 2-3	off	on 1-2	5	2	3	1 2/	1	2	25 µA 3/
-221		none	off							
-231			on 1-2							
-241		off	none							
-271	mom-on 2-3		mom-on 1-2							
-281	none									
-311	on 2-3									
-321 5/ 6/	none	on 2-3								

1/ Direction of movement of internal mechanism is opposite to the direction of the toggle movement.
2/ With time constant of .020 ±.002 second.
3/ Contact resistance not to exceed 50Ω during life, low current level switching.
4/ Non-functional terminals shall not be supplied.
5/ Dielectric withstanding voltage: 1,200 V rms at sea level (center on circuits).
6/ Delayed action of the switch toggle lever may cause circuit to close or open before snap action mechanism trips.

Referenced documents:
MIL-DTL-8834

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Review activities:
Navy - EC
Air Force - 99

Preparing activity:
DLA – CC

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